ANALYTICAL CUMULATIVE EXAM

November 16, 1996

1. You have been sent to India to investigate the occurrence of goiter disease attributed to iodine deficiency, As part of your investigation, you must make field measurements of traces of iodide in groundwater. The procedure is to oxidize I^- to I_2 and convert the product into an intensely colored complex with the dye brilliant green in the organic solvent toluene. Iodide and brilliant green form a 1:1 ratio. A 3.15 X 10-6 M solution of the colored complex exhibited an absorbance of 0.267 at 635 nm in a 1.00 cm cuvet. A blank solution made from distilled water in place of groundwater had an absorbance of 0.019. The absorbance of an unknown solution prepared from groundwater was 0.175. Find the concentration of iodide in the unknown solution.

2. The molar absorptivity of benzoic acid in methanol is about 1950 at 275 nm. What is the maximum concentration of benzoic acid in <u>g/mL</u> that can be used in a 1.00 cm cell if the absorbance is not to exceed 1.3?

3. A cadmium-zinc alloy is known to contain about 5% iron. It is proposed to analyze for the iron by specific complexing with 5-sulfonanthraniclic acid. As carried out, a 0.2 g sample is dissolved in dilute sulfuric acid, the pH adjusted, and complexing agent added. The final volume is 400 mL. At 455 nm this solution shows an absorbance of 0.637 in a 1.00 cm cell. Previously it was established that the molar absorptivity of the pure iron complex under the same conditions is 1306. A blank containing only complexing agent at the supposed excess concentration gave a absorbance of 0.015. Calculate the percentage of iron in the 0.2 g sample.